MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

THIRD SEMESTER

Discipline Specific Core Course (DSC)-MG3DSCCSC200- Database Management Systems

(2024 ADMISSION ONWARDS)

Duration : 1.5 Hours **Maximum Marks : 35**

1. Coding and Output - 20 Marks
2. Viva - 10 Marks
3. Record - 5 Marks

Q1. STUDENT TABLE

I. Create a table named STUDENT with the following structure:(5 Marks)

Column Name	Data Type	Constraints
enrollment_no	VARCHAR(10)	PRIMARY KEY
student_name	VARCHAR(50)	NOT NULL
department	VARCHAR(30)	NOT NULL
semester	INT	
cgpa	DECIMAL(3,2)	
city	VARCHAR(30)	

II. Insert the following records into the STUDENT table:(5 Marks)

enrollment_no	student_name	department	semseter	CGPA	city
CS001	Rahul Sharma	Computer Science	4	8.5	Delhi
CS002	Priya Singh	Computer Science	6	9.2	Mumbai
EC001	Amit Kumar	Electronics	4	7.8	Bangalore
ME001	Sneha Patel	Mechanical	2	8.0	Pune
CS003	Arjun Verma	Computer Science	4	6.5	Delhi
EC002	Divya Reddy	Electronics	6	9.5	Hyderabad
ME002	Rohan Gupta	Mechanical	4	7.2	Chennai
CS004	Kavya Nair	Computer Science	2	8.8	Kochi

III. Generate and execute the following queries

1. Add a new column phone_number of type VARCHAR(15) to the STUDENT table.(2 marks)
2. Update the CGPA of student 'Arjun Verma' (enrollment_no: CS003) to 7.0. (2marks)
3. Delete all students from the STUDENT table who have a CGPA less than 7.5. (2 marks)
4. Find the average CGPA of all students in the Computer Science department. (2 marks)
5. Display the number of students in each department. (2 marks)

Q2. Library Management System

I. Create a table BOOK with the following columns (5marks)

Column Name	Data Type	Constraints
book_id	VARCHAR(10)	PRIMARY KEY
title	VARCHAR(100)	NOT NULL
author	VARCHAR(50)	NOT NULL
genre	VARCHAR(30)	
price	DECIMAL(6,2)	
publisher	VARCHAR(50)	

II. Insert the following records to the table (5 Marks)

book_id	title	author	publisher	price
B001	Database Systems	Ramez Elmasri	Pearson	850.00
B002	Operating Systems	Abraham Silberschatz	Wiley	920.00
B003	Data Structures	Seymour Lipschutz	McGraw Hill	650.00
B004	Computer Networks	Andrew Tanenbaum	Pearson	780.00
B005	Python Programming	Mark Lutz	O'Reilly	1200.00
B006	Algorithms	Thomas Cormen	MIT Press	1500.00
B007	Web Development	Jon Duckett	Wiley	550.00
B008	Software Engineering	Roger Pressman	McGraw Hill	890.00

III. Generate and execute the following queries

1. Add a new column quantity to the table of type INT.(2 marks)
2. Modify the price of the book titled “Web Development” as 650.00.(2 marks)

3. Delete books with a price less than 300. (2 marks)
4. Find the average price of all books by publisher “Pearson”. (2 marks)
5. Display the details of the books in the order of price. (2 marks)

Q3.Employee Details

I. Create a table named EMPLOYEE with the following structure. (5 marks)

Column Name	Data Type	Constraints
emp_id	VARCHAR(10)	PRIMARY KEY
emp_name	VARCHAR(50)	NOT NULL
designation	VARCHAR(40)	
department	VARCHAR(30)	NOT NULL
salary	DECIMAL(10,2)	Should be greater than 0
joining_date	DATE	

II. Insert the following records into the EMPLOYEE table. (5 marks)

emp_id	emp_name	designation	department	salary	joining_date
E001	John Smith	Manager	Sales	65000.00	2020-03-15
E002	Sarah Johnson	Developer	IT	55000.00	2021-06-20
E003	Mike Brown	Analyst	Finance	48000.00	2019-08-10

E004	Emily Davis	Developer	IT	58000.00	2022-01-05
E005	David Wilson	Executive	Sales	42000.00	2021-11-12
E006	Lisa Anderson	Manager	Finance	70000.00	2018-05-25
E007	Tom Martinez	Developer	IT	52000.00	2023-02-18
E008	Anna Taylor	Executive	Sales	38000.00	2022-09-30

III. Generate and execute the following queries

1. Increase the size of department column to 35.(2 marks)
2. Update the salary of all employees in the IT department by increasing it by 10%.(2 marks)
3. Delete all employees from the EMPLOYEE table who have a salary less than 40000. (2 marks)
4. Find the maximum salary among all employees in the Sales department.(2 marks)
5. Display the average salary of employees in each department. (2 marks)

Q4. Inventory Management System

I. Create a table named PRODUCT with the following structure. (5marks)

Column Name	Data Type	Constraints
product_id	VARCHAR(10)	PRIMARY KEY
product_name	VARCHAR(60)	NOT NULL
category	VARCHAR(30)	NOT NULL
brand	VARCHAR(40)	
price	DECIMAL(10,2)	
stock_qty	INT	

II. Insert the following records into the PRODUCT table. (5 marks)

product_id	product_name	category	brand	price	stock_qty
P001	Laptop	Electronics	Dell	55000.00	25
P002	Smartphone	Electronics	Samsung	28000.00	40
P003	Washing Machine	Appliances	LG	32000.00	15
P004	Refrigerator	Appliances	Whirlpool	45000.00	12
P005	Headphones	Electronics	Sony	3500.00	60
P006	Microwave Oven	Appliances	Samsung	12000.00	20
P007	Tablet	Electronics	Apple	38000.00	18
P008	Air Conditioner	Appliances	Daikin	48000.00	8

III. Generate and execute the following queries

1. Add a new column warranty_period of type VARCHAR(20) to the PRODUCT table. (2 marks)
2. Update the price of all products in the Electronics category by reducing it by 5%.(2 marks)
3. Delete all products from the PRODUCT table where stock quantity is less than 10. (2 marks)
4. Display the products in Appliances category in the highest order of price.(2 marks)
5. Display the total stock quantity of products in each category. (2 marks)

Q5. Customer Order System

I. Create a table named ORDERS with the following structure. (5marks)

Column Name	Data Type	Constraints
order_id	VARCHAR(10)	PRIMARY KEY
customer_name	VARCHAR(50)	NOT NULL
product_name	VARCHAR(60)	NOT NULL
quantity	INT	
price_per_unit	DECIMAL(10,2)	

II. Insert the following data into the ORDERS table. (5marks)

order_id	customer_name	product_name	quantity	price_per_unit
O001	Rajesh Kumar	Laptop	2	52000.00
O002	Priya Sharma	Mobile Phone	1	25000.00
O003	Amit Patel	Tablet	3	18000.00
O004	Neha Singh	Headphones	5	2500.00

O005	Vikram Reddy	Smart Watch	2	15000.00
O006	Anjali Gupta	Laptop	1	52000.00
O007	Suresh Nair	Keyboard	4	1500.00
O008	Kavita Joshi	Mouse	6	800.00

III. Generate and execute the following queries

1. Change the size of customer_name to 55. (2marks)
2. Increase the quantity of products to 3 which are less than 3. (2marks)
3. Delete the product whose price is less than 1000. (2marks)
4. Find the total number of laptops ordered. (2marks)
5. Find the total quantity of each ordered product. (2marks)

Q6. Sports Club System

I. Create a table named MEMBER with the following structure. (5marks)

Column Name	Data Type	Constraints
member_id	VARCHAR(10)	PRIMARY KEY
member_name	VARCHAR(50)	NOT NULL
sport	VARCHAR(30)	NOT NULL
membership_type	VARCHAR(20)	
fees	DECIMAL(8,2)	
age	INT	
joining_date	DATE	

II. Insert the following data into the MEMBER table. (5marks)

member_id	member_name	sport	membership_type	fees	age	joining_date
MB001	Rohit Sharma	Cricket	Annual	15000.00	28	2023-01-15
MB002	Saina Nehwal	Badminton	Monthly	2500.00	25	2023-06-20
MB003	Virat Kohli	Cricket	Annual	15000.00	30	2022-08-10
MB004	Mary Kom	Boxing	Quarterly	8000.00	35	2023-03-25
MB005	Neeraj Chopra	Athletics	Annual	12000.00	26	2023-02-18
MB006	PV Sindhu	Badminton	Annual	14000.00	27	2022-11-05
MB007	Sunil Chhetri	Football	Monthly	3000.00	32	2023-07-12
MB008	Hima Das	Athletics	Quarterly	7500.00	24	2023-05-08

III. Generate and execute the following queries

1. Modify the size of sport to 35. (2marks)
2. Update the membership type of member 'Saina Nehwal' to 'Annual' and fees to 14000.00. (2marks)
3. Delete all members from the MEMBER table who have 'Monthly' membership type. (2marks)
4. Display the details of members with ‘Annual’ membership in the alphabetical order of their names. (2marks)
5. Find the total membership of each type. (2marks)

Q7. Hospital Management System

I. Create a table named PATIENT with the following structure. (5marks)

Column Name	Data Type	Constraints
patient_id	VARCHAR(10)	PRIMARY KEY
patient_name	VARCHAR(50)	NOT NULL
age	INT	
gender	VARCHAR(10)	
disease	VARCHAR(50)	
doctor_name	VARCHAR(50)	

II. Insert the following data into the PATIENT table. (5marks)

patient_id	patient_name	age	gender	disease	doctor_name
P101	Rajesh Mehta	45	Male	Diabetes	Dr. Sharma
P102	Anita Desai	32	Female	Fever	Dr. Patel
P103	Sunil Kapoor	58	Male	Heart Disease	Dr. Kumar
P104	Priya Nair	28	Female	Dengue	Dr. Sharma
P105	Mohan Lal	62	Male	Hypertension	Dr. Verma
P106	Seema Roy	40	Female	Arthritis	Dr. Kumar
P107	Arjun Singh	35	Male	Fever	Dr. Patel
P108	Kavita Jain	50	Female	Diabetes	Dr. Sharma

III. Generate and execute the following queries

1. Add a new column admission_date of date data type. (2marks)

2. Update the doctor name for patient 'Anita Desai' to 'Dr. Verma'. (2marks)
3. Display the details of patients above 50 years. (2marks)
4. Find the average age of patients in the table. (2marks)
5. Display the number of male and female patients. (2marks)

Q8. Movie Theater System

I. Create a table named MOVIE with the following structure. (5marks)

Column Name	Data Type	Constraints
movie_id	VARCHAR(10)	PRIMARY KEY
title	VARCHAR(80)	NOT NULL
genre	VARCHAR(30)	
director	VARCHAR(50)	
duration	INT	
rating	DECIMAL(2,1)	
release_date	DATE	

II. Insert the following data into the MOVIE table. (5marks)

movie_id	title	genre	director	duration	rating	release_date
M001	Inception	Sci-Fi	Christopher Nolan	148	8.8	2010-07-16
M002	The Shawshank Redemption	Drama	Frank Darabont	142	9.3	1994-09-23
M003	The Dark Knight	Action	Christopher Nolan	152	9.0	2008-07-18

M004	Forrest Gump	Drama	Robert Zemeckis	142	8.8	1994-07-06
M005	Interstellar	Sci-Fi	Christopher Nolan	169	8.6	2014-11-07
M006	The Godfather	Crime	Francis Coppola	175	9.2	1972-03-24
M007	Avatar	Sci-Fi	James Cameron	162	7.9	2009-12-18
M008	Titanic	Romance	James Cameron	194	7.9	1997-12-19

III. Generate and execute the following queries

1. Add a default value constraint to the column rating with value 4.0. (2marks)
2. Update the ratings which are less than 8 to 8.0. (2marks)
3. Display the details of movies in Sci-Fi genre. (2marks)
4. What is the average rating of movies in the table. (2marks)
5. Sort the details of movies in the highest order of rating. (2marks)

Q9. Course Registration System

I. Create a table named REGISTRATION with the following structure. (5marks).

Column Name	Data Type	Constraints
reg_id	VARCHAR(10)	PRIMARY KEY
student_name	VARCHAR(50)	NOT NULL
course_name	VARCHAR(60)	NOT NULL
course_code	VARCHAR(10)	
credits	INT	

semester	INT	
reg_date	DATE	

II. Insert the following data into the REGISTRATION table. (5marks)

reg_id	student_ name	course_name	course_code	credits	semester	reg_date
R001	Amit Sharma	Database Systems	CS301	4	3	2024-08-01
R002	Neha Gupta	Operating Systems	CS302	4	3	2024-08-02
R003	Rahul Verma	Data Structures	CS201	3	2	2024-08-01
R004	Priya Singh	Web Development	CS303	3	3	2024-08-03
R005	Vikram Reddy	Database Systems	CS301	4	3	2024-08-02
R006	Anjali Patel	Computer Networks	CS304	4	3	2024-08-04
R007	Suresh Kumar	Data Structures	CS201	3	2	2024-08-01
R008	Kavita Iyer	Machine Learning	CS401	4	4	2024-08-05

III. Generate and execute the following queries

1. Add a default value constraint to the column credits with value 3. (2marks)
2. Update the credits for course 'Web Development' to 4. (2marks)
3. Display the details of courses in semester 3 and credit 4. (2marks)
4. How many courses are there with credit 4. (2marks)

5. Find the total number of credits registered by student 'Amit Sharma'. (2marks)

Q10. Vehicle Showroom System

I. Create a table named VEHICLE with the following structure. (5marks).

Column Name	Data Type	Constraints
vehicle_id	VARCHAR(10)	PRIMARY KEY
model_name	VARCHAR(50)	NOT NULL
brand	VARCHAR(30)	NOT NULL
color	VARCHAR(20)	
price	DECIMAL(10,2)	
mfg_year	INT	
fuel_type	VARCHAR(15)	

II. Insert the following data into the VEHICLE table. (5marks)

vehicle_id	model_name	brand	color	price	mfg_year	fuel_type
V001	City	Honda	White	1250000.00	2023	Petrol
V002	Creta	Hyundai	Red	1580000.00	2024	Diesel
V003	Swift	Maruti	Blue	850000.00	2023	Petrol
V004	Seltos	Kia	Black	1680000.00	2024	Diesel
V005	Venue	Hyundai	Silver	1180000.00	2023	Petrol
V006	Baleno	Maruti	Grey	920000.00	2024	Petrol
V007	XUV700	Mahindra	White	2150000.00	2024	Diesel

V008	Nexon	Tata	Orange	1120000.00	2023	Electric

III. Generate and execute the following queries

1. Add a new column seating_capacity of type INT to the VEHICLE table.. (2marks)
2. Update the price of all vehicles with fuel type 'Electric' by increasing it by 8%. (2marks)
3. Delete all vehicles from the VEHICLE table manufactured in year 2023. (2marks)
4. Find the average price of all vehicles of brand 'Hyundai'.(2marks)
5. Display the number of vehicles available for each fuel type.(2marks)